

Personal Information

Name Saptarshi Pal
Email spal@math.harvard.edu
Others links Google Scholar, ResearchGate

Research Interests

a) Game Theory b) Human and AI Behavior c) Evolutionary Biology.

Teaching Interests

a) Game Theory b) Evolutionary Game Theory c) Evolutionary Dynamics d) Mathematical Biology.

Academics

March 2024– **Postdoctoral Fellow and Lecturer**,
Department of Mathematics, Harvard University.
2020–2024 **PhD in Mathematics and Natural Sciences**, *magna cum laude*,
Max Planck Institute for Evolutionary Biology.
Supervisor: Dr. Christian Hilbe
2018–2020 **Masters in Mathematics**,
University of Waterloo.
Supervisors: Dr. Chris T. Bauch and Dr. Madhur Anand
2014–2018 **Bachelor in Engineering**,
Jadavpur University, Kolkata, India.

Publications

- 2026 S. Pal, A. Mallela, L. Pracher, C. Wei, F. Fu, S. Schnell, M A Nowak.
Large Language Models instantiate evolutionarily robust strategies of cooperation, PNAS Nexus.
- 2025 Philip LaPorte, Lenz Pracher, Saptarshi Pal
From simultaneous to leader-follower play in direct reciprocity, PNAS Nexus
- 2025 Saptarshi Pal, Mayeul Lambert, Martin A. Nowak.
Stabilizing unconditional cooperation, Proceedings of Royal Society A.
- 2024 Saptarshi Pal, Nikoleta E. Glynatsi, Christian Hilbe.
The co-evolution of direct, indirect and generalized reciprocity, arXiv
- 2023 Marta C. Couto and Saptarshi Pal.
Introspection dynamics in asymmetric multiplayer games, Dynamic Games and Applications
- 2022 Saptarshi Pal and Christian Hilbe.
Reputation effects drive the joint evolution of cooperation and social rewarding, Nature Communications
- 2021 Saptarshi Pal, Chris T. Bauch, and Madhur Anand.
Coupled social and land use dynamics affect dietary choice and agricultural land-use extent
Communications Earth & Environment.

Teaching experience

- Fall 2026 Lecturer at Harvard University, *Math 242: Mathematical Models of Evolution*
Spring 2026 Lecturer at Harvard University, *Math 243: Evolutionary Dynamics*
Spring 2024 Lectures at Harvard University, *Math 243, Evolutionary Dynamics*.
2022 Lectures for graduate course in University of Kiel on *Evolutionary Game Theory*.
2022 Teaching Assistant for Math-WS21, University of Lübeck.
Mathematical Biology in action during the Corona pandemic
2018-2020 Teaching Assistant at Department of Mathematics, University of Waterloo
(linear algebra, multivariate calculus, mathematical biology)

Academic supervisions

- Fall 2024 **Lenz Pracher**. Visiting masters from LMU Munich at Harvard (incoming PhD student to MIT/Stanford).
Spring 2024 **Mayeul Lambert**. Visiting masters student from ENSTA Paris at Harvard.
Spring 2024 **Kevin Wu**. Bachelor's thesis at Harvard.
2024-2025 **Aarush Gunjal**. High school project at Newark Charter School, DE.

Professional activities

- 2021-Present **Peer Reviewing** at
Science, PNAS, PNAS Nexus, Nature Communications, Nature Sustainability, Dynamic Games and Applications, PLOS One.
2023 **Organizer** of lecture series on adaptive dynamics by Dr. Eva Kisdi at MPI Plön.

Scientific talks and visits

- 2026(Talk) Strategies of cooperation and defection in five large language models,
NITMB, Chicago, May, 2026.
2026(Talk) From simultaneous to leader-follower play in direct reciprocity,
AMS Sectional Meeting, Boston, March 2026.
2023(Talk) A model of plausible deniability,
Learning, Evolution and Games, Amsterdam.
2022(Talk) Title: Reputation facilitates the co-evolution of cooperation and social rewarding
EHBEA, Max Planck Institute for Evolutionary Anthropology, Leipzig
International Conference on Social Dilemmas, Copenhagen
2022 (School) *Toulouse Summer School in Quantitative Social Science* by IAST and Toulouse School of Economics
2022(School) *The Helsinki Summer School on Mathematical Ecology and Evolution*.
2017 (Intern) Max Planck Institute for Dynamics and Self Organization, Göttingen.
Supervisor: Prof. Marc Timme. (funded via the DAAD-WISE scholarship award)

Past collaborators

Christian Hilbe. Professor, ITU Austria, Linz.
Jörg Gross. Professor, University of Zürich.
Philip LaPorte. PhD student, UC Berkeley.
Marta Couto. Postdoctoral Fellow, University of Amsterdam.
Nikoleta E Glynnatsi. Research Scientist, RIKEN Japan.
Martin Nowak. Professor, Harvard University.

Programming skills

PYTHON, MATHEMATICA, MATLAB, $\text{\LaTeX}$